WEEKLY REPORT :

Project Title: Self Driving RC Car

Date: 11/3/2026

Instructor: Dr. Nguyen

TA: Ruben Ramirez

Project Members:

Ethan Wells (Group Leader)

Alexis Perez

Ethan Bishop

Abigail Duran

Weekly Summary (11/3/2026):

- This week the team tested every BuggyBot component together on the bench, finished the custom PCB design, and started the project poster and a ROS 2 version of the sign pipeline. Ethan Wells met both of last week's goals. The integration test passed all seven stages on the Raspberry Pi Compute Module 5 (CM5) I/O board, powered from the Pi with no motor load: the I2C scan found the LCD and IMU, both TCRT5000 line sensors responded, and the sign pipeline read a printed card at 98% confidence with 1.9 ms inference at 36 fps. The test found and fixed two bugs: the line sensor script was missing a required setting, and the I2C scan missed the LCD because the LCD only answers writes. He read the Cokoino 4WD motor HAT's schematic, found that it needs a 7-12 V input, and bought a 60 W adjustable bench supply. He then split the test into a per-component test and a stress test that runs the motors, camera and sign model, IMU, line sensors, and LCD all at once. On 9 V from the bench supply, the 120-second stress test passed with no errors and no under-voltage, and the camera held 30.0 fps against a 30.2 fps baseline. He also designed a 3D-printed bracket that holds the Camera Module 3 above the I/O board, and gave Abigail Duran final approval to send the PCB design for manufacturing. Alexis Perez worked on the Semester 1 project poster (due 11/5) instead of last week's goal of fine-tuning OpenCV: he collected the project information and results and designed the poster. Ethan Bishop partly met his goal of a first ROS 2 node on

the CM5: he wrote a preliminary ROS 2 version of the sign pipeline with camera, sign classifier, and LCD nodes. It runs in Docker on a development machine, because ROS 2 has no Raspberry Pi OS packages, and has not yet run on the CM5 or been added to the repository. He also reprinted the camera bracket taller (170 mm), wider, and with a larger brace, helped with the stress test, and supplied a battery to make the test setup mobile. Abigail Duran finished tracing the connections on the PCB and generated the manufacturing files: Gerbers, bill of materials, and pick-and-place.

Proposed Plan for Next Week:

- Ethan Wells: Write the Self-analysis, Project, and Technical Reports (due 11/19) and fine-tune OpenCV detection in the Pi sign demo.
- Alexis Perez: Plan how the vision model will be integrated with the vehicle.
- Ethan Bishop: Run the ROS 2 sign program on the CM5 with the camera on the new mount.
- Abigail Duran: Receive the PCB and begin testing it.

Weekly Contributions:

- Ethan Wells:
    - Ran the integration test on the CM5 I/O board: all seven stages passed; fixed the line sensor and LCD detection bugs it found
    - Read the Cokoino HAT schematic and bought a 60 W adjustable bench supply for its 7-12 V input
    - Split the test into test_components.py and stress_test.py; the 120-second stress test passed on 9 V with no errors or under-voltage at 30.0 fps
    - Designed a 3D-printed Camera Module 3 bracket for the CM5 I/O board
    - Gave Abigail Duran final approval to send the PCB design for manufacturing
    - Prepared the weekly slides and report
- Alexis Perez:
    - Collected project information and results for the Semester 1 poster
    - Designed the project poster
- Ethan Bishop:
    - Wrote a preliminary ROS 2 version of the sign pipeline (camera, sign classifier, and LCD nodes), running in Docker
    - Reprinted the camera bracket taller (170 mm), wider, and with a larger brace

    - Helped with the stress test and supplied a battery to make the test setup mobile
- Abigail Duran:
    - Finished tracing the connections on the PCB
    - Generated the manufacturing files: Gerbers, bill of materials, and pick-and-place

Hour Tracker:

Name	Hours For the Week	Cumulative Hours
Ethan Wells	6	64
Alexis Perez	6	54
Ethan Bishop	7	53
Abigail Duran	6	53